RBF-FD Solver: Least Squares Method for Operator Reproduction

Dennis Corraliza

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1 Problem Setting

Following the standard RBF-FD methods, let $u \in \mathcal{F} = W^{s,p}$ be a solution for a PDE in form of:

$$-Lu = -\nabla\cdot\big(A\nabla u\big) = f \quad \text{in } \Omega, \tag{1}$$

Throughout this document, we require that L be a self-adjointed, that is, $L = L^*$. Let $1 < p < \infty$, $A \in W^{1,\infty}(\bar{\Omega}; \mathbb{R}^{d\times d}) : \bar{\Omega} \to \mathbb{R}^{d\times d}$ is a symmetric positive definite anisotropic tensor, where

$$\zeta|\xi|^2 \leq \mathbf{A}(\mathbf{x})\xi\cdot\xi \leq \mathbf{Z}|\xi|^{\mathbf{2}},\ \mathbf{0} < \zeta \leq \mathbf{Z} \tag{2}$$

Assume $\Omega \subset \mathbb{R}^d$ and $f \in W^{s-k,p}(\Omega)$ is the forcing term acting in the domain Ω. For 2nd order elliptic operators, we set $k = 2$. The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within $\partial\Omega$ of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \tag{3}$$
$$n\cdot A\nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \tag{4}$$

For simplicity, we assume Ω be a bounded $C^{1,1}(\partial\Omega)$ domain, with $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$. Furthermore, let $g_D \in W^{s-1/p,p}(\partial\Omega)$, $g_N \in W^{s-1-1/p,p}(\partial\Omega)$. By *Agmon-Douglis-Nirenberg* Theorem [2], we have the following estimate for u:

$$\|u\|_{W^{s,p}(\Omega)} \leq C\left(\|Lu\|_{W^{s-k,p}(\Omega)} + \|g_D\|_{W^{s-1/p,p}(\partial\Omega)} + \|g_N\|_{W^{s-1-1/p,p}(\partial\Omega)} + \|u\|_{L^p(\Omega)}\right). \tag{5}$$

In the case that we work on a Hilbert space, we then have for $p = 2$ ($W^{s,2} := H^s$) the following estimate for u:

$$\|u\|_{H^s(\Omega)} \leq C\left(\|Lu\|_{H^{s-k}(\Omega)} + \|g_D\|_{H^{s-\frac{1}{2}}(\partial\Omega)} + \|g_N\|_{H^{s-\frac{3}{2}}(\partial\Omega)} + \|u\|_{L^2(\Omega)}\right).$$

If we want to study cases where $(Lu)(\boldsymbol{x})$ is known pointwise, we can formulate the setup by having $f \in W^{s-k,p}(\Omega)$ and, by Morrey Embedding,

$$W^{s-k,p}(\Omega) \hookrightarrow C^{\mu,\gamma}(\Omega) \hookrightarrow C^0(\Omega), \text{ with } \mu := \left\lfloor s-k-\frac{d}{p}\right\rfloor, \gamma := s-k-\mu-\frac{d}{p}$$

In other words, $s > k + \frac{d}{p}$. Suppose $p = 2$ and $k = 2$, then $s > 2 + \frac{d}{2}$. Given an operator $L : H^s(\Omega) \to H^{s-2}(\Omega)$, we define an error functional of the form:

$$\xi_{L,x} : H^s(\Omega) \to \mathbb{R},\ \xi_{L,x} \in (H^s)^*(\Omega)$$

Then by Riesz Representation, there exists a $v_{L,x} \in H^s(\Omega)$ such that $\xi_{L,x}(u) = \langle u, v_{L,x}\rangle_{H^s(\Omega)}$, and $\|\xi_{L,x}|\|_{(H^s(\Omega))^*} = \|v_{L,x}\|_{H^s(\Omega)}$

2 Functional Analysis Aspect

We examine the structural framework that governs the RBF-FD method. Let $\mathcal{N}(\Omega) := W^{s,p}(\Omega)$ be the native space whose vectors are point-wise bounded. We later assume that $\mathcal{N}(\Omega)$ is a Hilbert space (this holds, for instance, when $\mathcal{N}(\Omega)$ is the native space of a symmetric positive definite kernel). Let $L : \mathcal{N}(\Omega) \to W^{s-k,p}(\Omega)$ be a bounded linear operator. For $n \in \mathbb{N}$, let $\theta_1, \ldots, \theta_n \in \mathcal{L}\big(\mathcal{N}(\Omega), W^{s-k,p}(\Omega)\big)$ be bounded linear maps. Define $L_w : \mathcal{N}(\Omega) \to W^{s-k,p}(\Omega)$ by

$$L_w u := \sum_{i=1}^{n} \theta_i(u). \tag{6}$$

This operator is linear and bounded, since

$$\|L_w u\|_{W^{s-k,p}(\Omega)} \leq \|\boldsymbol{w}\|_{l^2} \, \|\theta\|_{\mathcal{L}\left(\mathcal{N}(\Omega), W^{s-k,p}(\Omega)\right)} \, \|u\|_{\mathcal{N}(\Omega)}.$$

The goal is for L_w to approximate L.

We further assume that there is a linear subspace $\mathcal{V}_h(\Omega) \subseteq \mathcal{N}(\Omega)$ on which L_w reproduces L exactly, i.e., $Lu = L_w u$ for all $u \in \mathcal{V}_w(\Omega)$. We define the error operator $\xi_{w,L} \in \mathcal{L}\big(\mathcal{N}(\Omega), W^{s-k,p}(\Omega)\big)$ by

$$\xi_{w,L}(u) := Lu - L_w u.$$

It is linear and bounded because L and L_w are, and it vanishes on $\mathcal{V}_w(\Omega)$. Let $1 < p < \infty$ and let q satisfy $\frac{1}{p} + \frac{1}{q} = 1$. For $g \in L^q(\Omega)$ define the linear functional l_g on $W^{s-k,p}(\Omega)$ by

$$l_g(v) := \sum_{|\alpha| \leq k} \int_\Omega v \, g \, d\boldsymbol{x}.$$

With $\|v\|_{W^{s-k,p}(\Omega)} = \left(\sum_{|\alpha|\leq k} \|D^\alpha v\|^p_{L^p(\Omega)}\right)^{1/p}$, the Hölder inequality (applied to the integrals and then to the finite sum over α) gives

$$|l_g(v)| \leq \|v\|_{W^{s-k,p}(\Omega)} \|g\|_{L^q(\Omega)}.$$

Set $\xi_{w,L,g} := l_g \circ \xi_{w,L}$. It is linear, and for every $u \in \mathcal{N}(\Omega)$

$$\begin{aligned} |\xi_{w,L,g}(u)| &\leq \|\xi_{w,L}(u)\|_{W^{s-k,p}(\Omega)} \|g\|_{L^q(\Omega)} \\ &\leq \|\xi_{w,L}\|_{\mathcal{L}(\mathcal{N}, W^{s-k,p})} \|g\|_{L^q(\Omega)} \|u\|_{\mathcal{N}(\Omega)}. \end{aligned}$$

Assuming $\xi_{w,L}$ is bounded, then $\xi_{w,L,g}$ is a bounded functional, $\xi_{w,L,g} \in \mathcal{N}(\Omega)^*$. Furthermore, if $\mathcal{N}(\Omega)$ is a Hilbert space, the Riesz Representation Theorem gives a unique $v_{w,L,g} \in \mathcal{N}(\Omega)$ such that $\xi_{w,L,g}(u) = \langle u, v_{w,L,g} \rangle_{\mathcal{N}(\Omega)}$ for all $u \in \mathcal{N}(\Omega)$. Moreover,

$$\begin{aligned} \|\xi_{w,L,g}\|_{\mathcal{N}^*(\Omega)} &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ \|u\|_{\mathcal{N}(\Omega)} \leq 1}} |\xi_{w,L,g}(u)| \\ &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ u \neq 0}} \frac{|\xi_{w,L,g}(u)|}{\|u\|_{\mathcal{N}(\Omega)}} \\ &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ u \neq 0}} \frac{|\langle u, v_{w,L,g} \rangle_{\mathcal{N}(\Omega)}|}{\|u\|_{\mathcal{N}(\Omega)}} = \|v_{w,L,g}\|_{\mathcal{N}(\Omega)}. \end{aligned}$$

2.1 Pointwise Evaluation

Suppose $s > k + \frac{d}{p}$. By Morrey Embedding, $W^{s-k,p}(\Omega) \hookrightarrow C^{\mu,\gamma}(\Omega) \hookrightarrow C^0(\Omega)$ Then we have a bounded functional $\delta_z \in \left(W^{s-k,p}(\Omega)\right)^*$:

$$\xi_{w,L}(u)(z) = \delta_z\left(\xi_{w,L}(u)\right) := \xi_{w,L,z}(u) \tag{7}$$

Then $\xi_{w,L,z} \in \mathcal{N}^*(\Omega)$ given $\xi_{w,L}$ is bounded. Furthermore, Assuming again $\mathcal{N}(\Omega)$ is Hilbert, then there exists a representer $v_{w,L,z}$ such that

$$\begin{aligned}
\|\xi_{w,L,z}\|_{\mathcal{N}(\Omega)^*} &= \sup_{\substack{u\in\mathcal{N}(\Omega)\\ \|u\|_{\mathcal{N}(\Omega)}\le 1}} |\xi_{w,L,z}(u)| \\
&= \sup_{\substack{u\in\mathcal{N}(\Omega)\\ u\neq 0}} \frac{|\xi_{w,L,z}(u)|}{\|u\|_{\mathcal{N}(\Omega)}} \\
&= \sup_{\substack{u\in\mathcal{N}(\Omega)\\ u\neq 0}} \frac{|\langle u, v_{w,L,z}\rangle_{\mathcal{N}(\Omega)}|}{\|u\|_{\mathcal{N}(\Omega)}} = \|v_{w,L,z}\|_{\mathcal{N}(\Omega)}.
\end{aligned}$$

We can concretely define $\xi_{w,L,z}$ as follows:

$$\xi_{w,L,z}(u) = Lu(z) - \sum_{i=1}^{n} \delta_z \circ \theta_i(u)$$

$$\xi_{w,L,z}(u) = Lu(z) - \sum_{i=1}^{n} w_{i,z} u(x_i)$$

Here we set $\delta_z \circ \theta_i = w_{i,z}\delta_{x_i}$, with $w_{1,z}, \ldots, w_{n,z} \in \mathbb{R}$.

3 RBF-FD Method

3.1 Local Approximation setup

To obtain an approximation of the operator L at a point $\boldsymbol{z} \in \Omega$, we approximate u by a function $\hat{u}$ on a neighborhood $\Omega_{\boldsymbol{z}} \subseteq \Omega$ of $\boldsymbol{z}$, so that $Lu(\boldsymbol{z}) \approx L\hat{u}(\boldsymbol{z})$. Let $Y = \{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_m\} \subseteq \Omega_{\boldsymbol{z}}$ be a set of centers, let $\phi_i : \Omega_{\boldsymbol{z}} \to \mathbb{R}$, $i = 1, \ldots, m$, be basis functions, and define the vector space $V_Y = \operatorname{span}\{\phi_1, \ldots, \phi_m\}$. Let $X = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\} \subseteq \Omega_{\boldsymbol{z}}$ be the set of stencil nodes for $\boldsymbol{z}$. For the classical approximation scheme, let $m = n$. We define $\boldsymbol{\Phi} \in \mathbb{R}^{n \times m}$ by $\boldsymbol{\Phi}_{ij} = \phi_j(\boldsymbol{x}_i)$ and assume that $\boldsymbol{\Phi}$ is invertible (for example, $\boldsymbol{y}_i = \boldsymbol{x}_i$ and a strictly positive definite kernel). The role of these basis functions will be to approximate the operator L. We seek $\hat{u} \in V_Y$ of the form

$$\hat{u}(\boldsymbol{x}) = \sum_{i=1}^{n} \lambda_i \phi_i(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{\lambda}, \qquad \boldsymbol{\phi}(\boldsymbol{x}) := [\phi_1(\boldsymbol{x}), \ldots, \phi_n(\boldsymbol{x})]^T, \tag{8}$$

that interpolates u on the stencil, $\hat{u}(\boldsymbol{x}_i) = u(\boldsymbol{x}_i)$ for $i = 1, \ldots, n$. This leads to the form $\boldsymbol{\Phi}\boldsymbol{\lambda} = \boldsymbol{u}$ for $\boldsymbol{u} = [u(\boldsymbol{x}_1), \ldots, u(\boldsymbol{x}_n)]^T$. We can solve for $\boldsymbol{\lambda}$ to obtain:

$$\hat{u}(\boldsymbol{x}) = \sum_{i=1}^{n} (\boldsymbol{\Phi}^{-1}\boldsymbol{u})_i \, \phi_i(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{\Phi}^{-1} \boldsymbol{u}.$$

Applying L (which acts on the argument $\boldsymbol{x}$ of $\boldsymbol{\phi}$) and evaluating at $\boldsymbol{z}$, the approximation of Lu at $\boldsymbol{z}$ is

$$Lu(\boldsymbol{z}) \approx L\hat{u}(\boldsymbol{z}) = (L\boldsymbol{\phi})(\boldsymbol{z})^T \boldsymbol{\lambda} = (L\boldsymbol{\phi})(\boldsymbol{z})^T \boldsymbol{\Phi}^{-1} \boldsymbol{u} =: \boldsymbol{w}^{L,X}(\boldsymbol{z})^T \boldsymbol{u}, \tag{9}$$

where $(L\boldsymbol{\phi})(\boldsymbol{z}) := [(L\phi_1)(\boldsymbol{z}), \ldots, (L\phi_n)(\boldsymbol{z})]^T$ and the weights $\boldsymbol{w}^{L,X}(\boldsymbol{z}) \in \mathbb{R}^n$ form a column vector.

Thus, approximating the operator reduces to a local linear solve for the weights. Transposing the definition of $\boldsymbol{w}^{L,X}(\boldsymbol{z})$ gives

$$\boldsymbol{\Phi}^T \boldsymbol{w}^{L,X}(\boldsymbol{z}) = (L\boldsymbol{\phi})(\boldsymbol{z}). \tag{10}$$

The method of undetermined coefficients on the basis $\{\phi_i\}$ gives the same weights: requiring exactness on each basis function, $\sum_{i=1}^{n} w_i \, \phi_j(\boldsymbol{x}_i) = (L\phi_j)(\boldsymbol{z})$ for $j = 1, \ldots, n$, is exactly the system above. Finally, we let the basis of V_Y be defined by radial basis functions, $\phi_i(\cdot) := \varphi(\|\cdot - \boldsymbol{y}_i\|)$, with a radial function $\varphi : [0, \infty) \to \mathbb{R}$ and centers $\{\boldsymbol{y}_i\}_{i=1}^m \subseteq \Omega_{\boldsymbol{z}}$ introduced above. Thus we have defined the Radial Basis Function–generated Finite Difference method (RBF-FD).

Remark. For $\boldsymbol{x} \in \Omega_{\boldsymbol{z}}$ let $w_{i,z} := \left(\boldsymbol{w}_z^{L,X}\right)_i$ be the weights computed with the fixed stencil X, and take $\theta_i = \delta_{\boldsymbol{x}_i}$. Then $L_w u = \sum_{i=1}^{n} w_i \, \theta_i(u) = L\hat{u}$ on $\Omega_{\boldsymbol{z}}$, that is, $L_w = L \circ I_X$ where $I_X u := \hat{u}$ is the interpolation operator. If $u \in V_Y$, then $I_X u = u$ and hence $L_w u = Lu$. The exactness assumption of the previous section holds with $\mathcal{V}_w = V_Y$ (working on $\Omega_{\boldsymbol{z}}$ in place of Ω and assuming $V_Y \subseteq \mathcal{N}$).

3.1.1 PD Consistency Error Analysis and Local System Derivation

We begin by defining the native space of solutions for (1) with (4). Suppose $\mathcal{F}_{\mathcal{K}} = H^s(\Omega)$ with pointwise bounded evaluations. This defines an RKHS for functions $u \in H^s(\Omega)$, $Lu \in H^{s-k}(\Omega)$. We can also define a *Kernel* $K : \Omega \times \Omega \to \mathbb{R}$ such that:

$$\begin{aligned} f(\boldsymbol{z}) &= \langle f, K(\cdot, \boldsymbol{z}) \rangle_{\mathcal{F}_{\mathcal{K}}} \\ (Lf)(\boldsymbol{z}) &= \langle f, L_x K(\cdot, \boldsymbol{z}) \rangle_{\mathcal{F}_{\mathcal{K}}} \\ \boldsymbol{w}_z^T \boldsymbol{f} &= \left\langle f, \boldsymbol{w}_z^T \boldsymbol{k}(\boldsymbol{\cdot}) \right\rangle_{\mathcal{F}_{\mathcal{K}}} \\ \xi_{w,L,z}(f) &:= Lf(z) - \boldsymbol{w}_{\boldsymbol{z}}^T \boldsymbol{f} \end{aligned}$$

According to [5], we have the following proposition:

Proposition 1. *(Davydov, 2025)Let D be a differential operator of order k, and K a cpd kernel of order s on $\Omega \subseteq \mathbb{R}^d$ and $\boldsymbol{X} = \{x_1, \dots, x_n\}$, assume that*

1. *$\partial^{\alpha,\beta} K(\boldsymbol{z}, \boldsymbol{z})$ exists for all $|\alpha|, |\beta| \leq k$*
2. *$L_z K(\boldsymbol{z}, \boldsymbol{x}_i)$ exists for all $i = 1, \dots, N$ and*
3. *The equation 16 is consistent (for cpd kernels - next section)*

Then the kernel-based numerical differentiation formula satisfies:

$$|\xi_{w^*,L,z}(f)| = \left| Lf(\boldsymbol{z}) - \sum_{i=1}^{N} w^*_{z,i} f(\boldsymbol{x_i}) \right| \leq \mathcal{P}_{K,L,X}(z) ||f||_{\mathcal{F}_\mathcal{K}} \tag{11}$$

We note that L is hence, a functional over $\mathcal{F}_\mathcal{K}$. L_z indicates applying the operator acted on the variable $\boldsymbol{z}$ in the kernel. We define $\boldsymbol{k}(\boldsymbol{z}) = [K(\boldsymbol{z}, \boldsymbol{x}_1), \dots, K(\boldsymbol{z}, \boldsymbol{x}_n)]^T$. Then $\xi_{w,L,z}(f) = \langle f, v_{w,L,z} \rangle$, with $v_{w,L,z} = L_z K(\cdot, \boldsymbol{z}) - \boldsymbol{w}_z^T \boldsymbol{k}(\cdot)$. By assuming point-wise error bounds, $\|\xi_{w,L,z}\|_{\mathcal{F}^*_\mathcal{K}}$, with error estimate $\|\xi_{w,L,z}\|_{\mathcal{F}^*_\mathcal{K}} = ||v_{w,K,L,X,z}||_{\mathcal{F}_\mathcal{K}}$. By [5], the functional is continuous on $\mathcal{F}_\mathcal{K}$ if $L_x K$ exists and both, polynomial consistency and operator exactness are guaranteed. Finally, We define the power function:

$$\mathcal{P}^2_{K,L,X}(z) = \min \left\{ ||v_{w,K,L,X,z}||^2_{\mathcal{F}_\mathcal{K}} : \boldsymbol{w} \in \mathbb{R}^n,\, \xi_{w,L,z} p = 0 \text{ for } p \in \Pi^d_{k-1} \right\} \tag{12}$$

The RKHS norm can be simplified as

$$\begin{aligned}
||v_{w,K,L,X,z}||^2_{\mathcal{F}_\mathcal{K}} &= \langle v_{w,K,L,X,z}, v_{w,K,L,X,z} \rangle_{\mathcal{F}_\mathcal{K}} \\
&= \xi_{w,L,y} \xi_{w,L,x} K(y,x)|_{y=z=x} \\
&= \left\langle L_z K(\cdot, \boldsymbol{z}) - \boldsymbol{w}_{\boldsymbol{z}}^T \boldsymbol{k}(\cdot), L_z K(\cdot, \boldsymbol{z}) - \boldsymbol{w}_{\boldsymbol{z}}^T \boldsymbol{k}(\cdot) \right\rangle_{\mathcal{F}_\mathcal{K}} \\
&= \langle L_z K(\cdot, \boldsymbol{z}), L_z K(\cdot, \boldsymbol{z}) \rangle_{\mathcal{F}_\mathcal{K}} - 2 \left\langle L_z K(\cdot, \boldsymbol{z}), \boldsymbol{w}_{\boldsymbol{z}}^T \boldsymbol{k}(\cdot) \right\rangle_{\mathcal{F}_\mathcal{K}} + \left\langle \boldsymbol{w}_{\boldsymbol{z}}^T \boldsymbol{k}(\cdot), \boldsymbol{w}_{\boldsymbol{z}}^T \boldsymbol{k}(\cdot) \right\rangle_{\mathcal{F}_\mathcal{K}} \\
&= L_y L_x K(y,x)|_{y=z=x} - 2 \sum_{i=1}^{n} w_{i,\boldsymbol{z}} L_z K(\boldsymbol{z}, \boldsymbol{x}_i) + \sum_{i,j=1}^{n} w_{i,\boldsymbol{z}} w_{i,\boldsymbol{z}} K(\boldsymbol{x}_i, \boldsymbol{x}_j) \\
&= 2 \left(\frac{1}{2} L_y L_x K(y,x)|_{y=z=x} - \sum_{i=1}^{n} w_{i,\boldsymbol{z}} L_z K(\boldsymbol{z}, \boldsymbol{x}_i) + \frac{1}{2} \sum_{i,j=1}^{n} w_{i,\boldsymbol{z}} w_{i,\boldsymbol{z}} K(\boldsymbol{x}_i, \boldsymbol{x}_j) \right)
\end{aligned}$$

Minimizing this norm for $w \in \mathbb{R}^n$ with polynomial constraints leads to Eq (10). An error estimate for functions in $\mathcal{F}_\mathcal{K}$ of the form:

$$\begin{aligned}
|\xi_{w,L,z}(f)| &\leq \|\xi_{w,L,z}\|_{\mathcal{F}^*_\mathcal{K}} ||f||_{\mathcal{F}_\mathcal{K}} \\
\inf_{\boldsymbol{w} \in \mathbb{R}^n} |\xi_{w,L,z}(f)| &\leq \inf_{\boldsymbol{w} \in \mathbb{R}^n} \|\xi_{w,L,z}\|_{\mathcal{F}^*_\mathcal{K}} ||f||_{\mathcal{F}_\mathcal{K}} \\
|\xi_{w^*,L,z}(f)| &\leq \inf_{\boldsymbol{w} \in \mathbb{R}^n} ||v_{w,K,L,X,z}||_{\mathcal{F}_\mathcal{K}} ||f||_{\mathcal{F}_\mathcal{K}} \\
|\xi_{w^*,L,z}(f)| &\leq \mathcal{P}_{K,L,X}(\boldsymbol{z}) ||f||_{\mathcal{F}_\mathcal{K}}
\end{aligned}$$

3.2 Polynomial Augmentation

For conditionally positive definite kernels, we augment them through the form:

$$\hat{u}(\mathbf{x}) = \sum_{\mathbf{j=1}}^{\mathbf{n}} \lambda_{\mathbf{i}} \phi_{\mathbf{i}}(\mathbf{x}) + \sum_{\mathbf{j=1}}^{\mathbf{l}} \beta_{\mathbf{j}} \mathbf{p}_{\mathbf{l}}(\mathbf{x}) \tag{13}$$

$$\sum_{i=1}^{n} \lambda_i p_j(\boldsymbol{x_i}) = 0, \forall i = 1, \ldots, n \tag{14}$$

We can generate a Gram matrix equation as follows:

$$\begin{bmatrix} \hat{\boldsymbol{u}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi} & \boldsymbol{P} \\ \boldsymbol{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\alpha} \end{bmatrix}$$

Hence, we can approximate the operator as below:

$$(L\hat{u})(\boldsymbol{z}) = \begin{bmatrix} (\boldsymbol{L\phi})(\boldsymbol{z}) \\ (\boldsymbol{Lp})(\boldsymbol{z}) \end{bmatrix}^T \begin{bmatrix} \boldsymbol{\Phi} & \boldsymbol{P} \\ \boldsymbol{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \hat{\boldsymbol{u}} \\ \mathbf{0} \end{bmatrix} := \begin{bmatrix} \boldsymbol{w}_z \\ \boldsymbol{\nu}_z \end{bmatrix}^T \begin{bmatrix} \hat{\boldsymbol{u}} \\ \mathbf{0} \end{bmatrix} = \boldsymbol{w}_z^T \hat{\boldsymbol{u}} \tag{15}$$

Thus, we get the local augmented operator reproduction scheme:

$$\begin{bmatrix} (\boldsymbol{L\phi})(\boldsymbol{z}) \\ (\boldsymbol{Lp})(\boldsymbol{z}) \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}^T & \boldsymbol{P} \\ \boldsymbol{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{w}_z \\ \boldsymbol{\nu}_z \end{bmatrix} \tag{16}$$

The polynomial constraint setup can be viewed as a Lagrange multiplier to a minimization scheme for the weights. By [6], given the basis for V_Y, $(L\boldsymbol{p})(\boldsymbol{z}) \in Range(\boldsymbol{P}^T) \iff$ Eq. (16) is uniquely solvable, for any set X. Additionally, by [6], Eq. (15) hold $\iff$ Eq. (16) holds. Note that depending on the the basis V_Y, we get the degree of $p \in \Pi_{k-1}^d$. Below are common RBFs and their respective cpd orders: This order provides us both, the degree of the polynomials $k-1$ and the

Table 1: Common radial basis functions and their minimal CPD order.

RBF	**Radial function** $\phi(r)$	**CPD order** k
Gaussian	$\exp(-\varepsilon^2 r^2)$	0
Inverse multiquadric	$(r^2+\varepsilon^2)^{-1/2}$	0
Multiquadric	$(r^2+\varepsilon^2)^{1/2}$	1
Linear	r	1
Cubic	r^3	2
Thin-plate spline	$r^2 \log r$	2
Quintic	r^5	3

number of terms:

$$l = \binom{d+k-1}{k-1}$$

As shown in [8], a polynomial augmentation can be added with a larger degree than required for the RBF, acting as a correction term while inheriting the non-oscillatory behavior of the RBF at the boundary $\partial\Omega$.

3.2.1 CPD Consistency Error Analysis and Local System Derivation

Due to the positive semidefiniteness of the cpd kernels, the polynomial augmentation plays the role of filling the nullspace the kernel does not span. We define the RKHS $\mathcal{F}_{\mathcal{K}c} = \mathcal{F}_{\mathcal{K}}/\Pi^d_{k-1}$, with the equivalence relation $u\ v \to u - v \in \Pi^d_{k-1}$. We define the semi-norm $|\cdot|_{\mathcal{F}_{\mathcal{K}}}$ as:

$$|u|_{\mathcal{F}_{\mathcal{K}}} := \|[u]\|_{\mathcal{F}_{\mathcal{K}}/\Pi^d_{k-1}}, \text{ with } [u] \in \mathcal{F}_{\mathcal{K}}/\Pi^d_{k-1} \tag{17}$$

Thus, $|p|_{\mathcal{F}_{\mathcal{K}}} = 0$ for $p \in \Pi^d_{k-1}$. Similar to the pd case, we develop the error estimate:

$$\begin{aligned}
\xi_{w,L,z}(f) &= Lf(z) - \sum_{i=1}^{n} w_{i,z} f(x_i) \\
&= L_z \langle f,\, K(\cdot, \boldsymbol{z})\rangle_{\mathcal{F}_{\mathcal{K}c}} - \sum_{i=1}^{n} w_{i,z} \langle f,\, K(\cdot, \boldsymbol{x}_i)\rangle_{\mathcal{F}_{\mathcal{K}c}} \\
&= \left\langle f,\, L_z K(\cdot, \boldsymbol{z}) - \sum_{i=1}^{n} w_{i,z} K(\cdot, \boldsymbol{x}_i) \right\rangle_{\mathcal{F}_{\mathcal{K}c}} \\
&= \langle f,\, v_{w,L,K,X,z}\rangle_{\mathcal{F}_{\mathcal{K}c}}
\end{aligned}$$

The main difference between the cpd and pd error estimates is the norm/semi-norm used for $v_{w,K,L,z} := v_{w,K,z}$. We then state the following:

$$\begin{aligned}
\|v_{w,K,z}\|^2_{\mathcal{F}_{\mathcal{K}c}} &= \langle v_{w,K,z},\, v_{w,K,z}\rangle_{\mathcal{F}_{\mathcal{K}c}} \\
\|v_{w,K,z}\|^2_{\mathcal{F}_{\mathcal{K}c}} &= \left\langle L_z K(\cdot, \boldsymbol{z}) - \sum_{i=1}^{n} w_{i,z} K(\cdot, \boldsymbol{x}_i),\, L_z K(\cdot, \boldsymbol{z}) - \sum_{i=1}^{n} w_{i,z} K(\cdot, \boldsymbol{x}_i) \right\rangle_{\mathcal{F}_{\mathcal{K}c}} \\
\|v_{w,K,z}\|^2_{\mathcal{F}_{\mathcal{K}c}} &= L_x L_y K(\boldsymbol{x}, \boldsymbol{y})|_{x=y=z} - 2\sum_{i=1}^{n} w_{i,z} K(\boldsymbol{x}_i, \boldsymbol{z}) + \sum_{i,j=1}^{n} w_{j,z} w_{i,z} K(\boldsymbol{x}_i, \boldsymbol{x}_j) \\
\|v_{w^*,K,z}\|^2_{\mathcal{F}_{\mathcal{K}c}} &= \inf_{\substack{\boldsymbol{w}\in\mathbb{R}^n \\ \xi(\Pi^d_{k-1})=0}} \left(L_x L_y K(\boldsymbol{x}, \boldsymbol{y})|_{x=y=z} - 2\sum_{i=1}^{n} w_{i,z} K(\boldsymbol{x}_i, \boldsymbol{z}) + \sum_{i,j=1}^{n} w_{j,z} w_{i,z} K(\boldsymbol{x}_i, \boldsymbol{x}_j) \right)
\end{aligned}$$

Minimizing this norm for $w \in \mathbb{R}^n$ with polynomial constraints leads to Eq (16). An error estimate for functions in $\mathcal{F}_{\mathcal{K}}$ of the form:

$$\begin{aligned}
\inf_{\substack{\boldsymbol{w}\in\mathbb{R}^n \\ \xi(\Pi^d_{k-1})=0}} |\xi_{w,L,z}(f)| &\leq \inf_{\substack{\boldsymbol{w}\in\mathbb{R}^n \\ \xi(\Pi^d_{k-1})=0}} ||v_{w,K,z}||_{\mathcal{F}_{\mathcal{K}c}} ||f||_{\mathcal{F}_{\mathcal{K}c}} \\
|\xi_{w^*,L,z}(f)| &\leq ||v_{w^*,K,z}||_{\mathcal{F}_{\mathcal{K}c}} ||f||_{\mathcal{F}_{\mathcal{K}c}}
\end{aligned}$$

3.3 Growth Function

To define the consistency order, we define the following stencil as the stencil set.

$$S_{\boldsymbol{z},X} := \bigcup_{i=1}^{N} [\boldsymbol{z}, \boldsymbol{x}_i]$$

Then for $U : S_{\boldsymbol{x},X} \times S_{\boldsymbol{x},X} \to \mathbb{R}$, we define a binary Höder continuity semi-norm in the form:

$$|U|_{\boldsymbol{z},X,\gamma} := \sup_{\substack{\boldsymbol{x},\boldsymbol{y} \in S_{\boldsymbol{z},X} \\ \boldsymbol{x} \neq \boldsymbol{z}, \boldsymbol{y} \neq \boldsymbol{z}}} \frac{|U(\boldsymbol{x},\boldsymbol{y}) - U(\boldsymbol{x},\boldsymbol{z}) - U(\boldsymbol{z},\boldsymbol{y}) + U(\boldsymbol{z},\boldsymbol{z})|}{\|\boldsymbol{x} - \boldsymbol{z}\|_2^{\gamma} \|\boldsymbol{y} - \boldsymbol{z}\|_2^{\gamma}}, 0 < \gamma, \leq 1$$

Then, for $r \in \mathbb{N}$, we get

$$|U|_{\boldsymbol{z},X,r+\gamma} := \frac{1}{(\gamma+1)^2 \dots (\gamma+r)^2} \left(\sum_{|\alpha|=r} \sum_{|\beta|=r} \binom{r}{\alpha} \binom{r}{\beta} |\partial^{\alpha,\beta} U|^2_{\boldsymbol{z},X,\gamma} \right)^{1/2},$$

with the following coefficient definitions: $\binom{r}{\alpha} = \frac{r!}{\alpha!}, \alpha! = \alpha_1! \dots \alpha_d!$. Growth function is defined as follows:

$$\rho_{k-1,D}(\boldsymbol{z}, X, \mu) := \sup \left\{ Lp(\boldsymbol{z}) : p \in \Pi^d_{k-1}, |p(\boldsymbol{x_j})| \leq \|\boldsymbol{x}_j - \boldsymbol{x}\|_2^{\mu}, j = 1, \dots, n) \right\} \tag{18}$$

By [9], the dual form of ρ takes the form

$$\rho_{k-1,D}(\boldsymbol{z}, Z, \|\cdot\|) := \inf \left\{ \|\boldsymbol{w}\| : \boldsymbol{w} \in \mathbb{R}^n, \xi_{w,L,z} p = 0, p \in \Pi^d_{k-1} \right\}$$

We finally have the following by [5]:

Theorem 1. *(Davydov, 2025) Let L be a differential operator of order k and K a cpd kernel of order s on Ω. Given $\boldsymbol{z} \in \Omega$ and $\boldsymbol{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, with $S_{\boldsymbol{z},\boldsymbol{X}} \subset \Omega$. Assume that 16 is consistent and, for some $r \geq k$ and $\gamma \in (0, 1]$, the kernel K possesses continuous partial derivatives $\partial^{\alpha,\beta} K(\boldsymbol{x}, \boldsymbol{y})$ for all $|\alpha|, |\beta| \leq r$ and all $\boldsymbol{x}, \boldsymbol{y} \in S_{\boldsymbol{z},\boldsymbol{X}}$, and $|K|_{\boldsymbol{z},\boldsymbol{X},r+\gamma} < \infty$. Define $q = \max\{s, r+1\}$. Then for $f \in \mathcal{F}_{\mathcal{K}}$ the kernel-based numerical differentiation satisfies the error bound:*

$$|\xi_{w^*,L,z}| = \left| Lf(\boldsymbol{z}) - \sum_{i=1}^{N} w^*_{z,i} f(\boldsymbol{x_i}) \right| \leq \rho_{q,L}(\boldsymbol{z}, \boldsymbol{X}, r+\gamma) |K|_{\boldsymbol{z},\boldsymbol{X},r+\gamma} ||f||_{\mathcal{F}_{\mathcal{K}}} \tag{19}$$

4 Local Least Square for RBF-FD Weights

4.1 Weight Constraint Setup

An idea relevent to a least square procedure within the RBFFD machinery was initially proposed in [15], targeting the interpolator function. We propose instead analyzing the least squares in the direction of the numerical differentiation weights which leads to a minimum norm in the approximate solutions. Hence, rather than assuming a form for the solution, our goal is to approximate instead the operator through weights dual to u sampled at nodes X. Suppose $m \geq n$ (for the case of $m < n$, refer to [1]), thus more center nodes than stencil nodes per stencil. Let $\phi_j(\boldsymbol{z}) = K(\boldsymbol{z}, \boldsymbol{y}_j)$. The following notation for the systems of equations will be used, where X is the stencil nodes and Y is the center nodes:

$$\begin{aligned} K_{XY} &= [K(\boldsymbol{x}_i, \boldsymbol{y}_j)]_{i,j} \\ K_{YX} &= [K(\boldsymbol{x}_i, \boldsymbol{y}_j)]_{j,i} \\ K_{XX} &= [K(\boldsymbol{x}_i, \boldsymbol{x}_j)]_{i,j} \\ K_{YY} &= [K(\boldsymbol{y}_i, \boldsymbol{y}_j)]_{i,j} \end{aligned}$$

Additionally, instead of assuming exactness of the form $L_z K(\boldsymbol{y}_j, \boldsymbol{z}) = \sum_{i=1}^n w_{i,\boldsymbol{z}} K(\boldsymbol{y}_j, \boldsymbol{x}_i)$ for $j = 1, \ldots, m$ as in the classical RBF-FD, we define a residual function:

$$\begin{aligned} r_{w_z}(\boldsymbol{y}_j) :&= L_z K(\boldsymbol{y_j}, \boldsymbol{z}) - \sum_{i=1}^n w_{i,\boldsymbol{z}} K(\boldsymbol{y}_j, \boldsymbol{y}_i) \\ r_{w_z}(Y) &= L_z K(Y, \boldsymbol{z}) - K_{YX} \boldsymbol{w_z}, \ \boldsymbol{w} \in \mathbb{R}^n \text{ and } \boldsymbol{z} \in \Omega. \end{aligned}$$

4.2 Consistency Error Analysis through Center-Based Kernel

Similar to the local interpolation scheme to derive the error functional, we begin by defining the kernel used for the reproduction scheme. Rather than using the interpolation kernel K, we define the following kernel:

$$\tilde{K}(\boldsymbol{z}_1, \boldsymbol{z}_2) := \int_{\Omega'} K(\boldsymbol{y}, \boldsymbol{z}_1) K(\boldsymbol{y}, \boldsymbol{z}_2) d\mu(\boldsymbol{y}) \tag{20}$$

Here we set μ to be a measure and $\Omega' \subseteq \mathbb{R}^d$. We can show this is a positive semi-definite kernel, and positive definite on the set of nodes X by showing that the set $\{K(\boldsymbol{x}_1, \cdot), \ldots, K(\boldsymbol{x}_n, \cdot))\}$ is linearly independent in $L^2(\Omega', \mu)$. (NEED TO SHOW)

$$\begin{aligned} \xi_{w,L,\tilde{K},X,z}(f) &= Lf(z) - \sum_{i=1}^n w_{i,z} f(x_i) \\ &= L_z \left\langle f, \tilde{K}(\cdot, \boldsymbol{z}) \right\rangle_{\mathcal{F}_{\tilde{\mathcal{K}}}} - \sum_{i=1}^n w_{i,z} \left\langle f, \tilde{K}(\cdot, \boldsymbol{x}_i) \right\rangle_{\mathcal{F}_{\tilde{\mathcal{K}}}} \\ &= \left\langle f, L_z \tilde{K}(\cdot, \boldsymbol{z}) - \sum_{i=1}^n w_{i,z} \tilde{K}(\cdot, \boldsymbol{x}_i) \right\rangle_{\mathcal{F}_{\tilde{\mathcal{K}}}} \\ &= \left\langle f, v_{w,L,\tilde{K},X,z} \right\rangle_{\mathcal{F}_{\tilde{\mathcal{K}}}} \end{aligned}$$

with $v_{w,L,\tilde{K},X,\Omega',z} := L_z \tilde{K}(\cdot, \boldsymbol{z}) - \sum_{i=1}^n w_{i,z} \tilde{K}(\cdot, \boldsymbol{x}_i)$. Then, in the same way as in the interpolation case, the goal is to minimize the Riesz representer $v_{w,L,\tilde{K},X,Y,z} := v_{w,\tilde{K},z}$. Since $\tilde{K}$ is a kernel,

standard interpolation theory applies to get error estimates. However, rather than working with established theory, we proceed by exploiting the kernel structure.

$$
\begin{aligned}
\|v_{w,\tilde{K},z}\|^2_{\mathcal{F}_{\tilde{\mathcal{K}}}} &= \left\langle v_{w,\tilde{K},z}, v_{w,\tilde{K},z} \right\rangle_{\mathcal{F}_{\tilde{\mathcal{K}}}} \\
&= L_x L_y \tilde{K}(\boldsymbol{x},\boldsymbol{y})\Big|_{x=y=z} - 2\sum_{i=1}^{n} w_{i,z} L_z \tilde{K}(\boldsymbol{z},\boldsymbol{x}_i) + \sum_{i,j=1}^{n} w_{i,z} w_{j,z} \tilde{K}(\boldsymbol{x}_i,\boldsymbol{x}_j) \\
&= \int_{\Omega'} LzK(\boldsymbol{y},\boldsymbol{z}) L_z K(\boldsymbol{y},\boldsymbol{z})\, d\mu(\boldsymbol{y}) - \int_{\Omega'} 2\sum_{i=1}^{n} w_{i,z} L_z K(\boldsymbol{y},\boldsymbol{z}) K(\boldsymbol{y},\boldsymbol{x}_i)\, d\mu(\boldsymbol{y}) \\
&\quad + \int_{\Omega'} \sum_{i,j=1}^{n} w_{i,z} w_{j,z} K(\boldsymbol{y},\boldsymbol{x}_i) K(\boldsymbol{y},\boldsymbol{x}_j)\, d\mu(\boldsymbol{y}) \\
&= \int_{\Omega'} \left(LzK(\boldsymbol{y},\boldsymbol{z}) - \sum_{i=1}^{n} w_{i,z} K(\boldsymbol{y},\boldsymbol{x}_i) \right)^2 d\mu(\boldsymbol{y}) \\
&= \int_{\Omega'} r^2_{w_z}(\boldsymbol{y})\, d\mu(\boldsymbol{y}) \\
\inf_{\boldsymbol{w}\in\mathbb{R}^n} \|v_{w,\tilde{K},z}\|^2_{\mathcal{F}_{\tilde{\mathcal{K}}}} &= \inf_{\boldsymbol{w}\in\mathbb{R}^n} \int_{\Omega'} r^2_{w_z}(\boldsymbol{y})\, d\mu(\boldsymbol{y}) \\
\|v_{w^*,\tilde{K},z}\|_{\mathcal{F}_{\tilde{\mathcal{K}}}} &= \inf_{\boldsymbol{w}\in\mathbb{R}^n} \|r_{w_z}\|_{L^2(\Omega',\mu)}
\end{aligned}
$$

Assuming that K is a kernel for an RKHS $\mathcal{F}_{\mathcal{K}}$, if $T(g) := \int_{\Omega'} K(\boldsymbol{y},\cdot) g(\boldsymbol{y})\, d\mu(\boldsymbol{y})$, then:

$$
\begin{aligned}
\|v_{w,\tilde{K},z}\|_{\mathcal{F}_{\mathcal{K}}} &= \left\| L_z \tilde{K}(\cdot,\boldsymbol{z}) - \sum_{i=1}^{n} w_{i,z} \tilde{K}(\cdot,\boldsymbol{x}_i) \right\|_{\mathcal{F}_{\mathcal{K}}} \\
&= \left\| \int_{\Omega'} K(\boldsymbol{y},\cdot) \left(L_z K(\boldsymbol{y},\boldsymbol{z}) - \sum_{i=1}^{n} w_{i,z} K(\boldsymbol{y},\boldsymbol{x}_i) \right) d\mu(\boldsymbol{y}) \right\|_{\mathcal{F}_{\mathcal{K}}} \\
&= \left\| \int_{\Omega'} K(\cdot,\boldsymbol{y}) r_{w_z}(\boldsymbol{y})\, d\mu(\boldsymbol{y}) \right\|_{\mathcal{F}_{\mathcal{K}}} \\
&= \inf_{g: T(g) = T(r_{w_z})} \|g\|_{L^2(\Omega',\mu)} \\
&\leq \|r_{w_z}\|_{L^2(\Omega',\mu)} \\
&\leq \left(\int_{\Omega'} |r_{w_z}(\boldsymbol{y})|^2\, d\mu(\boldsymbol{y}) \right)^{1/2} \\
&\leq \left(\int_{\Omega'} |v_{w,K,z}(\boldsymbol{y})|^2\, d\mu(\boldsymbol{y}) \right)^{1/2} \\
&\leq \left(\int_{\Omega'} \|v_{w,K,z}\|^2_{\mathcal{F}_{\mathcal{K}}} \|K(\cdot,\boldsymbol{y})\|^2_{\mathcal{F}_{\mathcal{K}}}\, d\mu(\boldsymbol{y}) \right)^{1/2} \\
&\leq \left(\int_{\Omega'} \|K(\cdot,\boldsymbol{y})\|^2_{\mathcal{F}_{\mathcal{K}}}\, d\mu(\boldsymbol{y}) \right)^{1/2} \|v_{w,K,z}\|_{\mathcal{F}_{\mathcal{K}}}
\end{aligned}
$$

Finally, for a uniform, normalized measure and a translation-invariant kernel, then $\|v_{w,\tilde{K},z}\|_{\mathcal{F}_{\mathcal{K}}} \leq \|v_{w,K,z}\|_{\mathcal{F}_{\mathcal{K}}}$.

4.3 Consistency Error Derivation through Standard RBF-FD

We use the solution of the interpolation scheme to compare with the ordinary least squares, where $K_{XX}\boldsymbol{w}_z^* = L_z K(Y,\boldsymbol{z}) := \boldsymbol{b}$ is the local interpolation scheme. Thus, we can say $\boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* = \boldsymbol{b}^T K_{XX}^{-1}\boldsymbol{b}$ since K_{XX} is symmetric. Similar to the interpolation scheme, we define the error analysis and have the form:

$$\begin{aligned}
||v_{w,K,L,X,z}||^2_{\mathcal{F}_\mathcal{K}} &= L_y L_z K(y,z)|_{y=z=x} - 2\sum_{i=1}^{n} w_{i,z} L_z K(\boldsymbol{z},\boldsymbol{x}_i) + \sum_{i,j=1}^{n} w_{i,z} w_{j,z} K(\boldsymbol{x}_i,\boldsymbol{x}_j) \\
&= L_y L_z K(y,z) - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} \\
&= L_y L_z K(y,z) - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* - \boldsymbol{b}^T K_{XX}^{-1}\boldsymbol{b} \\
&= L_y L_z K(y,z) - \boldsymbol{b}^T\boldsymbol{w}^* - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* \\
&= L_y L_z K(y,z) - 2\boldsymbol{b}^T\boldsymbol{w}^* + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* \\
&= \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) - \boldsymbol{w}^T K_{XX}\boldsymbol{w}^* - \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* \\
&= \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) + (\boldsymbol{w}-\boldsymbol{w}^*)^T K_{XX}(\boldsymbol{w}-\boldsymbol{w}^*)
\end{aligned}$$

We can now bound the second term by considering $\boldsymbol{w} \to \boldsymbol{w_{LS}}$ obtained from a least squares result:

$$\begin{aligned}
K_{XY}K_{YX}\boldsymbol{w_{LS}} &= K_{XY}\boldsymbol{b} \\
K_{XY}K_{YX}\boldsymbol{w_{LS}} - K_{XY}K_{YX}\boldsymbol{w}^* &= K_{XY}\boldsymbol{b} - K_{XY}K_{YX}\boldsymbol{w^*} \\
K_{XY}K_{YX}(\boldsymbol{w_{LS}} - \boldsymbol{w}^*) &= K_{XY}(\boldsymbol{b} - K_{YX}\boldsymbol{w}^*) \\
(\boldsymbol{w_{LS}} - \boldsymbol{w}^*) &= K_{YX}^{\dagger}\boldsymbol{r}^* \\
||\boldsymbol{w_{LS}} - \boldsymbol{w}^*||_2 &\le ||K_{YX}^{\dagger}||_2 ||\boldsymbol{r}^*||_2 \\
||\boldsymbol{w_{LS}} - \boldsymbol{w}^*||_2 &\le \frac{||\boldsymbol{r}^*||_2}{\sigma_{min}(K_{XY})}
\end{aligned}$$

Finally, we say that $r_j^* = v_{w^*,K,L,X,z}(\boldsymbol{y}_j)$. Then $||\boldsymbol{r}^*||_2^2 = ||v_{w^*,K,L,X,z}(Y)||_2^2$. Applying kernel evaluation on $v_{w^*,K,L,X,z}$ and the power function for interpolations onto v we get:

$$\begin{aligned}
\|\boldsymbol{r}^*\|_2^2 &= \sum_{i=1}^{m} v^2_{w^*,K,L,X,z}(\boldsymbol{y}_i) \\
&\le \sum_{i=1}^{m} \|v_{w^*,K,L,X,z}\|^2_{\mathcal{F}_\mathcal{K}} \|K(\cdot,\boldsymbol{y}_i)\|^2_{\mathcal{F}_\mathcal{K}} \\
&\le \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) \sum_{i=1}^{m} K^2(\boldsymbol{y}_i,\boldsymbol{y}_i)
\end{aligned}$$

Therefore, the error estimate (21) for the local Ordinary Least Squares behaves as follows:

$$\begin{aligned}
||v_{w_{LS},K,L,X,z}||^2_{\mathcal{F}_\mathcal{K}} &\le \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) + \lambda_{max}(K_{XX})||\boldsymbol{w}-\boldsymbol{w}^*||_2^2 \\
&\le \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) + \lambda_{max}(K_{XX}) \frac{\mathcal{P}^2_{K,L,X}(\boldsymbol{z}) \sum_{i=1}^{m} K^2(\boldsymbol{y}_i,\boldsymbol{y}_i)}{\sigma^2_{min}(K_{XY})} \\
&\le \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) \left(1 + \frac{\lambda_{max}(K_{XX}) \sum_{i=1}^{m} K^2(\boldsymbol{y}_i,\boldsymbol{y}_i)}{\sigma^2_{min}(K_{XY})}\right)
\end{aligned} \tag{21}$$

4.4 Least Squares Finite Spaces and Interpolation Results

Suppose $\{p_i\}_{i=1}^l$ is a set of polynomials augmenting the radial basis functions. Let $\Pi_k = span\,\{p_i\}_{i=1}^l$ with polynomial degree up to k. We want to reproduce the operator exactly on the polynomial basis:

$$\begin{aligned} Lp_j(\boldsymbol{z}) &:= \sum_{i=1}^{n} w_{i,z} p_j(\boldsymbol{x_i}) \text{ for } j = 1, \dots, l \\ L\boldsymbol{p}(\boldsymbol{z}) &= \boldsymbol{P^T} \boldsymbol{w}_z \end{aligned}$$

where $\boldsymbol{P}_{ij} := p_j(\boldsymbol{x}_i) \in \mathbb{R}^{n \times l}$. Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\boldsymbol{w}_z) := \min \dfrac{1}{2} \|\boldsymbol{r}(\boldsymbol{w}_z)\|_2^2 \quad \text{for } \boldsymbol{x} \in \Omega \\ \text{subject to } K_{YX} \boldsymbol{w}_z = L\boldsymbol{p}(\boldsymbol{z}) \end{cases} \tag{22}$$

We can solve this setup by defining the Lagrangian $\mathcal{L}$ for the problem as follows:

$$\begin{aligned} \mathcal{L}(\boldsymbol{w}, \boldsymbol{\nu}) &:= J(\boldsymbol{w}) + \left\langle \boldsymbol{\nu},\, \boldsymbol{P}^T \boldsymbol{w} - L\boldsymbol{p}(\boldsymbol{z}) \right\rangle \\ &= \frac{1}{2} \left\langle K_{YX} \boldsymbol{w},\, K_{YX} \boldsymbol{w} \right\rangle - \left\langle K_{YX} \boldsymbol{w},\, L_z K(Y, \boldsymbol{z}) \right\rangle + \frac{1}{2} \left\langle L_z K(Y, \boldsymbol{z}),\, L_z K(Y, \boldsymbol{z}) \right\rangle \\ &\quad + \left\langle \boldsymbol{\nu},\, \boldsymbol{P}^T \boldsymbol{w} - L\boldsymbol{p}(\boldsymbol{z}) \right\rangle. \end{aligned}$$

The resulting KKT equations for this system are then:

$$\begin{aligned} \nabla_w \mathcal{L} = 0 &: K_{XY} L_z K(Y, \boldsymbol{z}) = K_{XY} K_{YX} \boldsymbol{w} + \boldsymbol{P}\boldsymbol{\nu} \\ \nabla_\nu \mathcal{L} = 0 &: \boldsymbol{P}^T \boldsymbol{w} = L\boldsymbol{p} \end{aligned}$$

The augmented matrix then takes the form:

$$\begin{bmatrix} K_{XY} K_{YX} & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} X_{XY} L_z K(Y, \boldsymbol{z}) \\ L\boldsymbol{p}(\boldsymbol{z}) \end{bmatrix} \tag{23}$$

From [6], Equation (23) has a unique solution given $L\boldsymbol{p} \in Range(\mathbf{P^T})$. The primary goal is to approximate $Lu(\boldsymbol{z}) \approx \boldsymbol{w}_z^T \boldsymbol{u}$, with $\boldsymbol{u} = [u(\boldsymbol{x}_1), \dots, u(\boldsymbol{x}_n)]^T$. Define $L_w f(\boldsymbol{z}) := \boldsymbol{w}_z^T \boldsymbol{f}$ for some function f. Thus, we have the following:

$$\begin{aligned} L_w f(\boldsymbol{z}) &= \boldsymbol{w}_z^T \boldsymbol{f} \\ &= \begin{bmatrix} \boldsymbol{w}_z^T & \boldsymbol{\nu}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{0} \end{bmatrix} \\ &= \left(\begin{bmatrix} K_{XY} K_{YX} & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix}^{-1} \begin{bmatrix} K_{XY} L_z K(Y, \boldsymbol{z}) \\ L\boldsymbol{p}(\boldsymbol{z}) \end{bmatrix} \right)^T \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{0} \end{bmatrix} \\ &= \begin{bmatrix} L_z K(\boldsymbol{z}, Y) K_{XY} & L\boldsymbol{p}^T(\boldsymbol{z}) \end{bmatrix} \begin{bmatrix} K_{XY} K_{YX} & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix}^{-T} \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{0} \end{bmatrix} \\ &= \begin{bmatrix} L_z K(\boldsymbol{z}, Y) K_{XY} & L\boldsymbol{p}^T(\boldsymbol{z}) \end{bmatrix} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{\alpha} \end{bmatrix} \\ &= L_z K(\boldsymbol{z}, Y) \left(K_{YX} \boldsymbol{q} \right) + L\boldsymbol{p}^T(\boldsymbol{z}) \boldsymbol{\alpha} \end{aligned}$$

Next, we can show that the coefficients $\boldsymbol{q}, \boldsymbol{\alpha}$ relate to the coefficients of u as follows:

$$\begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{0} \end{bmatrix} = \begin{bmatrix} K_{XY}K_{YX} & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{\alpha} \end{bmatrix}$$

Define $V_m = \{\rho(z) : \rho(\boldsymbol{z}) = \sum_{i=1}^{m} \lambda_i K(\boldsymbol{z}, \boldsymbol{y}_i) \text{ and } \lambda \in Range\,(K_{YX})\,, \boldsymbol{y}_i \in Y\} \subseteq V_Y$. Let $s \in V_m + \Pi_k$. Thus, $s(\boldsymbol{z}) = K(\boldsymbol{z}, Y)K_{XY}q + \boldsymbol{p}^T(\boldsymbol{z})\boldsymbol{\alpha}$ where $\boldsymbol{q} \in \mathbb{R}^n$ and $\boldsymbol{\alpha} \in \mathbb{R}^l$. Then:

$$\begin{aligned} \boldsymbol{s} = s(X) = K_{XY}K_{YX}\boldsymbol{q} + \boldsymbol{P}\boldsymbol{\alpha} = f(X) = \boldsymbol{f} \\ \boldsymbol{P}^T\boldsymbol{q} = \boldsymbol{0} \end{aligned}$$

Hence, we get interpolation of f at the node set X. Furthermore, from the KKT matrix we can show that $K_{XY}\boldsymbol{r} = \boldsymbol{P}\boldsymbol{\nu}$ which implies that $K_{XY}\boldsymbol{r} \in range(\boldsymbol{P}) = \left(ker\left(\boldsymbol{P}^T\right)\right)^{\perp}$. Thus, we get the relationship:

$$\boldsymbol{r}^T\left(K_{YX}\boldsymbol{q}\right) = \boldsymbol{0} \text{ for } \boldsymbol{P}^T\boldsymbol{q} = \boldsymbol{0} \tag{24}$$

Finally, we check the consistency error for $s \in V_h + \Pi_k$ with $\boldsymbol{P}^T\boldsymbol{q} = \boldsymbol{0}$.

$$\begin{aligned} Ls(\boldsymbol{z}) - L_h s(\boldsymbol{z}) &= L\left[K(\boldsymbol{z}, Y)K_YX\boldsymbol{q} + \boldsymbol{p}^T(\boldsymbol{z})\boldsymbol{\alpha}\right] - \boldsymbol{w}_z^T\boldsymbol{s} \\ &= L_z K(\boldsymbol{z}, Y)K_{YX}\boldsymbol{q} + L\boldsymbol{p}^T(\boldsymbol{z})\boldsymbol{\alpha} - \boldsymbol{w}_z^T(K_{XY}K_{YX}\boldsymbol{q} + \boldsymbol{P}\boldsymbol{\alpha}) \\ &= L_z K(\boldsymbol{z}, Y)K_{YX}\boldsymbol{q} + L\boldsymbol{p}^T(\boldsymbol{z})\boldsymbol{\alpha} - \boldsymbol{w}_z^T K_{XY}K_{YX}\boldsymbol{q} - \boldsymbol{w}_z^T\boldsymbol{P}\boldsymbol{\alpha} \\ &= \left[L_z K(\boldsymbol{z}, Y) - K_{YX}\boldsymbol{w}_z\right]^T K_{YX}\boldsymbol{q} + \left[L\boldsymbol{p}(\boldsymbol{z}) - \boldsymbol{P}^T\boldsymbol{w}_z\right]^T\boldsymbol{\alpha} \\ &= \boldsymbol{r}^T K_{YX}\boldsymbol{q} + \left[L\boldsymbol{p}(\boldsymbol{z}) - \boldsymbol{P}^T\boldsymbol{w}_z\right]^T\boldsymbol{\alpha} \end{aligned}$$

Hence, we conclude $Ls(\boldsymbol{z}) - L_h s(\boldsymbol{z}) = 0$ due to (23) and (24).

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5 Appendix